



Lab Task(s) Report – Lab 11

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Syllabus:

Database Systems Lab

Syllabus Code: CL-2005

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Instructor:

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**Task-01: Create a table “Books” with fields like ISBN_NO, Title, and Author.
Create an html form to insert data into using PHP., and display the data using an html table.**

Database Structure – Backend Setup:

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'library_db' selected. The main panel shows the 'Table structure' for the 'books' table. The table structure is as follows:

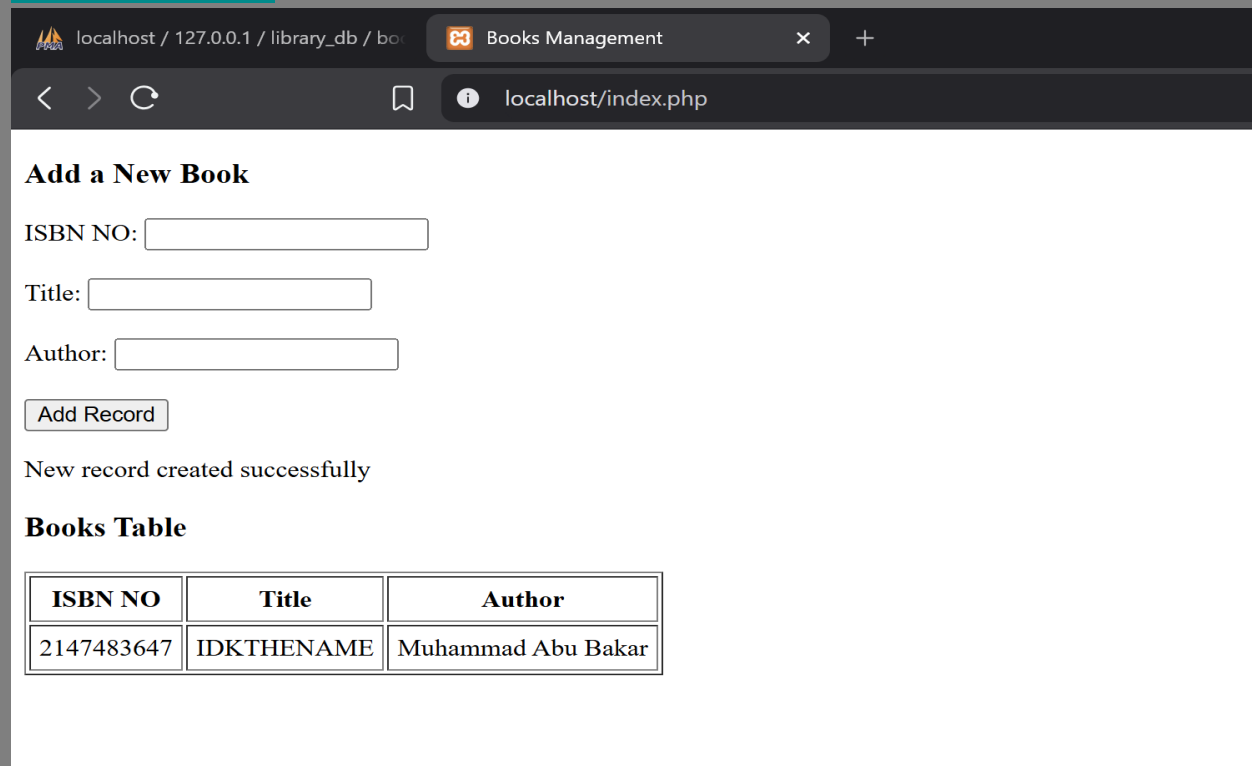
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	ISBN	int(11)			No	None			Change Drop More
2	Title	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
3	Author	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More

Below the table structure, there are options to 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Full'. There is also a 'Remove from central columns' button. At the bottom, there are buttons for 'Print', 'Propose table structure', 'Track table', 'Move columns', and 'Normalize'. A 'Go' button is also present.

The Frontend UI:

The screenshot shows the frontend UI for the Books Management system. The page has a title bar that says 'Books Management'. Below the title bar is a navigation bar with the URL 'localhost/index.php'. The main content area has a heading 'Add a New Book'. Below the heading is a form with three input fields: 'ISBN NO:', 'Title:', and 'Author:'. Below the form is a button labeled 'Add Record'. Below the button is a heading 'Books Table'. Below the heading is a table with three columns: 'ISBN NO', 'Title', and 'Author'.

Proof Of Insertion:



Add a New Book

ISBN NO:

Title:

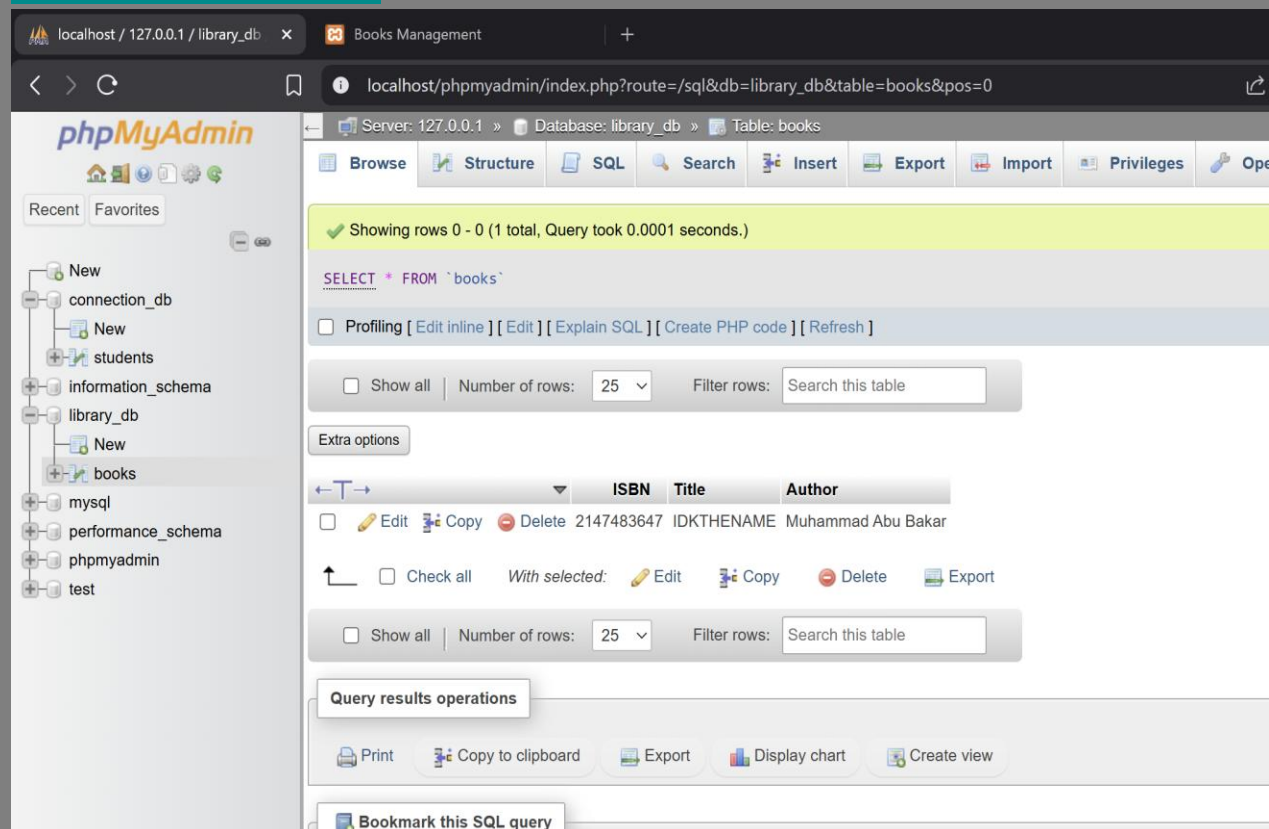
Author:

New record created successfully

Books Table

ISBN NO	Title	Author
2147483647	IDKTHENAME	Muhammad Abu Bakar

The Database Verification:



localhost / 127.0.0.1 / library_db Books Management

localhost/phpmyadmin/index.php?route=/sql&db=library_db&table=books&pos=0

Server: 127.0.0.1 » Database: library_db » Table: books

Browse Structure SQL Search Insert Export Import Privileges Open

Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)

`SELECT * FROM `books``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	ISBN	Title	Author
☐ Edit ☐ Copy ☐ Delete	2147483647	IDKTHENAME	Muhammad Abu Bakar

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

[Bookmark this SQL query](#)

Task-02: Replicate the same using C# with SQL Server using Windows Form OR using Django(Python).

```
mysql> USE library_db;
Database changed
mysql>
mysql> CREATE TABLE Books (
->     ISBN_NO VARCHAR(20) PRIMARY KEY,
->     Title VARCHAR(255) NOT NULL,
->     Author VARCHAR(255) NOT NULL
-> );
Query OK, 0 rows affected (0.07 sec)
```

Task-03: Perform the above task with Java using MYSQL in Visual Studio Code.)

```
package com.example.library;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;

@Controller
public class BookController {
    @Autowired
    private BookRepository repository;

    @GetMapping("/")
    public String index(Model model) {
        model.addAttribute("books", repository.findAll());
        model.addAttribute("book", new Book());
        return "index";
    }

    @PostMapping("/add")
    public String addBook(@ModelAttribute Book book) {
        repository.save(book);
        return "redirect:/";
    }
}
```



```
1 package com.example.library;
2 import org.springframework.data.jpa.repository.JpaRepository;
3
4 public interface BookRepository extends JpaRepository<Book, String> { }
```

```
1 package com.example.library;
2
3 import jakarta.persistence.Entity;
4 import jakarta.persistence.Id;
5
6 @Entity
7 public class Book {
8     @Id
9     private String isbnNo;
10    private String title;
11    private String author;
12
13    public String getIsbnNo() { return isbnNo; }
14    public void setIsbnNo(String isbnNo) { this.isbnNo = isbnNo; }
15    public String getTitle() { return title; }
16    public void setTitle(String title) { this.title = title; }
17    public String getAuthor() { return author; }
18    public void setAuthor(String author) { this.author = author; }
19 }
```

```

1  <!DOCTYPE html>
2  <html xmlns:th="http://www.thymeleaf.org">
3  <head>
4      <title>Java Library</title>
5      <style>
6          body { font-family: sans-serif; padding: 20px; }
7          table { width: 100%; border-collapse: collapse; margin-top: 20px; }
8          th, td { border: 1px solid #ddd; padding: 10px; }
9          form { background: #f9f9f9; padding: 15px; border-radius: 5px; }
10     </style>
11 </head>
12 <body>
13     <h2>Add Book (Java Spring)</h2>
14     <form th:action="@{/add}" th:object="${book}" method="post">
15         <input type="text" th:field="*{isbnNo}" placeholder="ISBN" required>
16         <input type="text" th:field="*{title}" placeholder="Title" required>
17         <input type="text" th:field="*{author}" placeholder="Author" required>
18         <button type="submit">Save Book</button>
19     </form>
20
21     <h2>Book List</h2>
22     <table>
23         <tr><th>ISBN</th><th>Title</th><th>Author</th></tr>
24         <tr th:each="b : ${books}">
25             <td th:text="${b.isbnNo}"></td>
26             <td th:text="${b.title}"></td>
27             <td th:text="${b.author}"></td>
28         </tr>
29     </table>
30 </body>
31 </html>

```

Add Book (Java Spring)

ISBN	Title	Author	Save Book
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Book List

ISBN	Title	Author
1234	abcd	Hammod
2345	Hello	Ali